Should You Pay to Remember? A Two-Gate, Pre-Registered Protocol for Deciding When Regime Retention Pays in Trading Systems

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Abstract

Regime-aware trading systems carry an evaluation problem that is worse than overfitting: their evaluations cannot fail. A regime-conditioned learner happily trains, "specializes," and beats a weak baseline on structureless labels, so a backtest win licenses nothing. We propose and field-test a pre-deployment protocol built around one number: the *retention deficit* $\hat{\Gamma}$, the post-reactivation decision-regret difference between the deployed always-retrain learner and an identical twin whose per-regime parameters are protected during dormancy—a two-arm counterfactual any desk can run before buying retention machinery. The protocol pairs $\hat{\Gamma}$ with a structure screen (regime-conditioning versus block-shuffled labels on the decision metric), which its own record demotes: across our field tests the screen's verdict dissociated from realized value three times, including its strongest pass coinciding with an inverted outcome. The field record spans daily crypto and commodities (screen fails, evaluations go flat, as required), 4-hour crypto (screen passes, $\hat{\Gamma} = 0$, flat again), and daily U.S. industry portfolios 1926–2025, where $\hat{\Gamma}$'s sign predicted the outcome direction in both directions: a positive deficit ($+0.00089 \pm 0.00011$) coincides with the full pre-registered arm ordering, a negative one with an inversion, and a frozen-specification replication on a withheld 64-year era scores 5 of 6 cells. Crisis-window transaction costs amplify rather than kill the deficit ($+0.00126 \to +0.00230$ as costs sweep 25–100 bps). An externally lodged, announcement-lagged NBER-recession test—registered with the OSF before recession dates ever touched the panel—resolved as a hit, with its two-recession concentration disclosed. A break-even rule converts $\hat{\Gamma}$ into a dollar-denominated carrying-cost threshold. All claims are regret and retention claims, not alpha claims; a live temporal test (diagnostics frozen 2026-06-30, scored 2027-04-07) and a constituent-level replication are lodged in advance.

CCS Concepts

• **Applied computing** → **Economics**; • **Computing methodologies** → *Machine learning*.

Keywords

regime switching, continual learning, decision-focused learning, evaluation protocols, pre-registration, transaction costs

1 Introduction: the evaluation that cannot fail

Every multi-strategy desk faces the same quiet question: the crisis playbook has been idle for six years—who is paying to keep it calibrated, and is that money buying anything? The machine-learning literature now offers an inventory of retention machinery to answer "yes": per-regime expert libraries, replay buffers, parameter isolation, continual-learning regularizers ported to finance [16, 28, 29, 31, 33]. What it does not offer is a way to find out that the answer is "no."

The reason is structural. A regime-aware evaluation has more researcher degrees of freedom than an ordinary backtest—the labeler, its thresholds, the dormancy definition, the evaluation windows—and its machinery produces organized-looking behavior on any labels at all: a regime-conditioned learner trains, converges, and "specializes" whether or not the labels carry information. Beating a generic baseline is therefore not evidence. *A method whose evaluation cannot go flat on structureless data should not be trusted when it goes sharp.* This paper is organized around engineering evaluations that can go flat, running them on real substrates, and reporting the flats, the inversions, and the hits in one ledger.

We work in the register of decision regret, not P&L: the per-day gap between the portfolio chosen from a model's predictions and the portfolio an oracle would have chosen, the predict-then-optimize lesson that prediction error and decision quality dissociate [8, 22]. The question the protocol prices is narrow and universal: when a dormant market regime returns, does a capacity-bounded learner that spent the dormancy retraining on whatever was active pay a measurable penalty in the first days of the reactivation—the days when decisions matter most—relative to a twin that kept the dormant regime's parameters untouched? If yes, retention is worth money and the break-even arithmetic says how much; if no, every dollar of retention machinery is premium paid on a claim that will never be filed. Recent theory makes the same point from first principles: a bounded agent should not retain what it can cheaply relearn [17]; our contribution is the measurement that decides the question per substrate, before deployment.

Contributions. (1) *A two-gate pre-deployment protocol* for regime-aware trading systems: a cheap structure screen run first, a binding deficit diagnostic run second, and a break-even carrying-cost rule that converts the diagnostic into dollars (Section 2). (2) *The diagnostic itself*: $\hat{\Gamma}$, a two-arm counterfactual (deployed learner versus its dormancy-protected twin) measured in post-reactivation decision regret—one number, causal by construction, cheap enough to pre-register and run on any book. (3) *A resolved field record* (Section 3): seven substrates spanning crypto, commodities, and a century of daily U.S. equity industry data, every null and inversion included, with the screen's honest demotion, a costs battery in which the deficit strengthens net of crisis-window costs (Section 4), an externally custodied forward test that has since resolved, and a live temporal test whose diagnostics are already frozen (Section 5). The retention mechanism, its theory, and the controlled dissociation studies are developed in a companion manuscript [2] [TODO:

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arxiv-id]; this paper is the protocol, the number, and the record a practitioner needs to decide whether to pay to remember.

1.1 Position among existing practice

Three literatures meet at this question and each leaves it unpriced. *Regime-switching finance*—Markov-switching models [12], regime effects in returns [1], regime-conditioned allocation [11, 26, 27], regime-weighted forecast combination [7]—conditions a model or its combination weights on an inferred state, implicitly assuming every component remains fully estimated while its state is dormant. That assumption is exactly what $\hat{\Gamma}$ measures the failure of under bounded capacity. *Continual learning in finance* makes the opposite unconditional bet: retention machinery is deployed because forgetting is presumed harmful [16, 28, 29, 31, 33], with evaluation by averaged forgetting metrics [21] rather than by a pre-deployment test of whether the substrate offers anything to retain. Our record contains real substrates where that bet loses ($\hat{\Gamma} < 0$ with a realized inversion), which is why the protocol runs *before* the purchase. *The backtest-overfitting protocol literature*—deflated Sharpe ratios [3], SPA and Reality-Check family tests [13, 34], the broader discipline of [14, 20]—is our closest methodological ancestor: it converts an unfalsifiable claim ("my strategy works") into a testable one by charging for selection. The two-gate protocol does the same for a different claim ("my system needs to remember"), and inherits that literature's stance: the deliverable is the test, and the test must be able to say no.

2 The protocol

2.1 Decision regret and probe windows

A stream of trading periods presents per-asset features X_t and next-period returns y_t, with a causal regime label r_t computed from information through $t - 1$ only. The decision layer is a cardinality-constrained long-only portfolio (top-k by predicted return, position cap $w_{\max}$), and the per-period *decision regret* of a prediction $\hat{y}$ is $\rho(\hat{y}; y) = y^\top z^*(y) - y^\top z(\hat{y}) \geq 0$: the return given up relative to the oracle portfolio. Regret, not prediction error, is the protocol's metric throughout—errors that do not change the selected book are free, and errors at the selection boundary are not [8].

Regimes recur with dormancy: the crisis label may sleep for years. The protocol's unit of account is the *probe window*: the first $N_p = 15$ trading days of every regime block whose regime was dormant at least 90 days. Post-reactivation regret—mean daily regret inside probe windows—is the primary metric because the reactivation transient is where retention could pay and where a fresh learner is most exposed; regime-conditioned finance has long known returns behave differently across states [1, 11, 12, 27], but conditions a model on the state without asking who maintained the model through the state's absence [7].

Conventions, fixed once. Every comparison in this paper is walk-forward with causal features; every claim is a pre-registered ordering prediction tested across seeds (100 where stated, else 20) with Welch tests and Holm–Bonferroni correction inside each registered pair family; paired designs expose every arm to identical data streams per seed. Where a design's seeds vary only initialization—the walk-forward batteries—the resulting intervals are implementation noise conditional on one market history, and every table says so. Refuted predictions are reported at the same prominence as confirmed ones; this paper contains both.

2.2 The structure screen, honestly demoted

Screen (structure). Fit one model per regime label, walk-forward; compare its decision regret against the identical machinery run on block-shuffled labels (≥ 50 draws), summarized as a z-score against the shuffle distribution (the empirical permutation p cannot fall below $1/51$; a normal fit to the control distribution gives the indicative p), with a split-half stability check and a comparison against a pooled unconditioned model. Block-shuffling preserves label persistence while destroying the label-return association, so the control inherits the machinery's ability to look organized on nothing. The screen asks only: *do these labels carry decision-relevant information at all?* It costs minutes and requires no retention machinery.

We designed the screen as a gate—no pass, no claims—and its own field record demoted it. Across the substrates of Section 3 the screen's verdict dissociated from realized retention value three times: a passing screen preceded an inverted outcome (daily equities under a volatility-band labeler), the strongest pass we ever measured ($z = 23.5$, withheld 1926–1989 era) coincided with an inversion, and a failing screen preceded a positive, cost-robust deficit (the NBER labeler). The screen therefore *licenses nothing*. It survives in the protocol for its original falsification role—where it fails, a null evaluation is the predicted outcome and a sharp one is suspect—and as a cheap negative control: machinery that "wins" where the screen fails is fitting noise.

2.3 The deficit $\hat{\Gamma}$: one number from two arms

Diagnostic (binding). Run two arms of identical capacity, features, and learning rates over the same walk-forward stream:

- **A1 (deployed default)**: a capacity-bounded learner that always trains on the currently active regime—the honest model of any allocation policy that follows current reward, whether a retraining schedule, a learned router, or recent-performance capital weighting;
- **A9 (protected twin)**: the same learner with per-regime parameters pinned—each regime's parameters are trained only on that regime's data and are untouched during its dormancy.

The *retention deficit* is $\hat{\Gamma} = \overline{\rho}_{\mathrm{A1}} - \overline{\rho}_{\mathrm{A9}}$ over probe windows: what dormancy-period retraining actually cost, in regret per probe day. In continual-learning terms $\hat{\Gamma}$ is a decision-regret backward-transfer measure against a multi-head oracle [21]; in econometric terms it descends from the post-break estimation-window question—how much pre-break knowledge should a forecaster retain [30]. It is causal by construction (the arms differ in exactly one design bit), cheap (two arms, no mechanism), and signed: $\hat{\Gamma} \leq 0$ is a finding, not a failure. Where $\hat{\Gamma}$ is positive, retention—however implemented—collects it; where it is not, no retention mechanism can be bought, because there is nothing to collect.

Both arms are stand-ins for classes, which is what makes the two-arm run sufficient. A1 represents every allocation policy whose training resource follows current reward—a retraining schedule, a learned router over experts, recent-performance capital weighting across desks—all of which starve the dormant regime identically, because a dormant regime emits no reward to follow. A9 represents every *isolation-preserving* implementation: a siloed desk, a frozen per-regime adapter store, an expert library with pinned members. The protocol prices the class boundary, not any implementation; choosing an implementation is a separate decision the companion studies [2]. Two practical notes. First, $\hat{\Gamma}$ is measured net of *rehearsals*: if near-miss episodes keep the dormant specialist warm between major events, that recalibration is in the measurement, which is the economically relevant accounting. Second, $\hat{\Gamma}$ should be re-measured net of transaction costs before pricing (Section 4); in our record costs move the number, in one case across zero.

2.4 Pricing the deficit: the break-even rule

$\hat{\Gamma}$ is in return units per probe day, so it converts directly to dollars. Charging a carrying cost c (dollars per idle trading day: infrastructure, data, attention, the calibration operation) against a sleeve of notional value V, retention of a regime whose dormancy is D trading days pays over one dormancy–reactivation cycle iff

$$\boxed{c \cdot D \;\leq\; \hat{\Gamma}_{\text{net}} \cdot N_p \cdot V}$$

where $\hat{\Gamma}_{\text{net}}$ is the deficit measured net of transaction costs (Section 4) and N_p is the probe length. The rule is a special case of the break-even analysis developed in the companion [2]; here it is used only as arithmetic.

Worked example (a \$100M crisis sleeve). Take the measured net-of-cost deficit at the 25 bps crisis-cost tier, $\hat{\Gamma}_{\text{net}} = +0.00126$ per probe day (Section 4): on \$100M that is \$126,000 per probe day, or ≈\$1.89M of avoidable regret per reactivation ($N_p = 15$). The June-2020 recession-cell reactivation in our record arrived after $D = 2{,}445$ trading days of dormancy; the rule prices retention as paying iff the sleeve's carrying cost is below \$1.89M/2,445 ≈ \$770 per idle trading day—about \$195,000 per year. A fully staffed idle desk at \$1M/year does not pay on this deficit; a frozen parameter store with a nightly calibration job at \$50K/year pays roughly fourfold. The rule's two-sided honesty is the point: it prices some retention out, and that is the protocol working.

3 The field record

Table 1 is the paper's spine: every substrate the protocol has touched, with the screen verdict, the measured deficit, and what the ten-arm evaluation machine (the companion's arm family, spanning reward-driven and retaining allocations [2]) actually did. Conventions: all labels causal; all comparisons walk-forward; Welch tests with Holm correction inside each pre-registered family; 100 seeds where stated, else 20. One caveat governs every real-data row and is repeated in Section 6.2: each walk-forward window is a single market history, so intervals are seed noise conditional on that history, not sampling uncertainty across markets.

Daily crypto and commodities: the protocol's floor. Neither small daily panel shows decision-metric structure under either causal labeler (all $|z| < 0.7$): regime-conditioned regret is statistically identical to block-shuffled-label regret. The ten-arm evaluation run on regime-stitched crypto schedules goes flat exactly as the failed screen predicts (no pair survives Holm correction; min Holm $p = 0.17$), and the numerically best arm is the one that *learns nothing*—a fixed-strategy combination—which sharpens the null: where labels carry no structure, machinery that adapts to them buys less than machinery that ignores them. This is the "can go flat" property demonstrated, not asserted.

4-hour crypto: structure without a deficit. A richer intraday feature inventory locates the program's first screen pass (951 episodes; conditioning beats shuffled labels by 3.15%, $z = 2.32$, both halves positive)—with its selection caveats stated at the same prominence: it is one confirmed cell from a ten-cell screen, and the confirmation reused the screen's data span, so the gap estimate is biased upward by selection on the maximum. The evaluation is flat anyway—and the two-arm counterfactual explains why retrospectively: $\hat{\Gamma} = +0.000002$, zero to four decimals. Reactivations are elevated for every arm including the protected twin; regime onsets are intrinsically hard here, but nothing is forgotten—at this bar density a fresh learner refits within the probe window. We are explicit about the epistemic order: the pre-registered prediction for this cell was separation, it was refuted, and the zero deficit is the diagnosis made after the refutation; the diagnosis's honest test is every substrate scored since. This cell is why the deficit, not the screen, is the binding diagnostic: structure is necessary for anything to be learnable, and irrelevant to whether retention pays.

Daily equities, 1990–2025: the deficit exists, and its sign prices the pool. On the Ken French 49 industry portfolios (daily, value-weighted, 1990–2025; 9,067 days), with designs and predictions pre-registered before each run, two labelers deliver the two halves of the story. Under the volatility-band labeler (L1) the screen passes and the deficit is *negative* (-0.00013 ± 0.00007): the machine run anyway produces a significant inversion—the always-retrain baseline beats the retaining arms in the probe window (Holm $1.2{\times}10^{-26}$), retention premium paid with no claim to collect. Under an index-drawdown labeler (L3, 15% threshold fixed in the registration before any run) the screen *fails* by the letter of its pooled sub-criterion while the deficit is positive—the first positive $\hat{\Gamma}$ on any real substrate in this program ($+0.00089 \pm 0.00011$, 100 seeds)—and the full pre-registered arm ordering emerges (joint Holm $5.2{\times}10^{-65}$), with the retaining arms also ahead on overall regret. One labeler-anatomy fact matters for reading the number: the *union* of the drawdown labeler's crisis states recurs every one to three years (the 2010, 2011, 2015–16, and 2018 near-misses are all in the inventory), while the specific crisis-with-uptrend cell sleeps up to 2,953 trading days—so retained parameters do receive live-fire recalibration between major crises, and $\hat{\Gamma}$ is measured net of those rehearsals, as a desk's economics would be. The deficit's evaluated inventory is 27 reactivations; under leave-one-reactivation-out it survives every single-event exclusion (range $[0.00071, 0.00101]$, all significant), survives excluding all of calendar-2020 ($+0.00080 \pm 0.00019$), and is independently positive within the 2010s ($+0.00077 \pm 0.00022$, n=14) and 2020s ($+0.00094 \pm 0.00040$, n=12).

Three disclosures belong beside that result. First, *threshold fragility*: the deficit lives at essentially one drawdown threshold per era—at 10/12/20% on 1990–2025 it is null or negative ($+0.00013 \pm 0.00027$, -0.00010 ± 0.00017, -0.00045 ± 0.00026)—and the pre-registered robustness prediction was refuted. Second, a *favorable wrinkle we flag because it was not predicted*: at 100 seeds the schedule-resampled (stitched) counterparts, previously flat, turn marginally positive in $\hat{\Gamma}$ under both labelers ($+0.00023 \pm 0.00016$ and $+0.00023 \pm 0.00015$)

Table 1: The field record: substrate × (structure screen, deficit $\hat{\Gamma}$, evaluation outcome). $\hat{\Gamma}$ is in decision-regret units per probe day (A1–A9, probe 15 days, dormancy ≥ 90 days; ± is a 95% CI over seeds, conditional on each window's single history). "Ordering" = the pre-registered arm ordering (retaining arms ahead of reward-driven arms); "inversion" = the deployed always-retrain baseline wins; "predates gate 2" = the battery ran before the deficit diagnostic entered the protocol. Every row's result files are released.

Substrate (span; labeler)	Structure screen	Deficit $\hat{\Gamma}$ (A1–A9)	What the machine did
Crypto, daily (5 pairs, 2021–25; vol/trend + filtered-vol)	fail ($z = -0.66$ / -0.17)	— (predates gate 2)	flat, as predicted (min Holm $p = 0.17$; 100 seeds)
Commodities, daily (3 series, 2015–25; both labelers)	fail ($z = -0.14$ / $+0.16$)	— (predates gate 2)	no dissociation run; claims forbidden by screen
Crypto, 4-hour (bar-native filtered-vol; confirmed cell)	pass (+3.15%, $z = 2.32$)	+0.000002 (zero to 4 dp)	flat ($p > 0.57$ all pairs): structure, nothing to retain
French 49 industries, daily (1990–2025; L1 vol-band)	pass (+1.14%, $z = 3.18$)	−0.00013 ± 0.00007 (100 s)	*inversion*: A1 beats pool (Holm $1.2{\times}10^{-26}$); reverses net of costs
French 49 industries, daily (1990–2025; L3 drawdown@15%)	fail by pooled letter ($z = 2.24$)	+0.00089 ± 0.00011 (100 s)	full ordering (joint Holm $5.2{\times}10^{-65}$); cost-amplified
Withheld era: 36 industries (1926–89; frozen L3 spec, 6 cells)	strongest pass ($z = 23.5$)	sign varies by cell	sign rule 5/6, both directions; screen's best pass → inversion
NBER recession × trend, announce-lagged (1990–2025)	fail ($z = 1.74$)	+0.00081 ± 0.00033	ordering (Holm $1.6{\times}10^{-4}$); two-recession concentration disclosed

while their arm families stay flat—consistent either with single-history optimism in the walk-forward precision or with stitching destroying exactly the multi-year dormancy structure that makes retention pay; the lodged constituent-level test discriminates. Third, *mechanism honesty*: registered probes found the deficit does not decay across the probe window and follows no dormancy or rehearsal covariate; it behaves as a persistent regime-conditional allocation deficit—the reward-following learner serves crisis windows persistently worse, the learner-side analogue of slow-moving capital [6, 24]—not demonstrated eviction-forgetting, which remains a controlled-setting result in the companion [2]. The deployment consequence (the counterfactual prices retention either way) is unchanged; the mechanism attribution is open, and we say so.

The withheld era, 1926–1989. The byte-identical frozen specification—same features, arms, tests, and the lodged sign hypothesis; zero new researcher degrees of freedom—was replicated on 36 industries over 16,983 days the design had never seen, in the pre-discovery-sample tradition [9, 18]. The screen delivers its strongest pass ever ($z = 23.5$) while the frozen 15% cell *inverts* ($\hat{\Gamma} = -0.00024 \pm 0.00008$; the always-retrain baseline wins, Holm $2.1{\times}10^{-3}$)—the dissociation that finally demoted the screen. The deficit's sign, by contrast, scored: of six withheld cells (thresholds 10/12/15/20%; two sub-eras at 15%), five are consistent with the lodged sign rule, in both directions, including two positive-deficit cells reproducing the full ordering out-of-sample (10%: $\hat{\Gamma} = +0.00020 \pm 0.00009$, Holm $1.9{\times}10^{-6}$; 1958–89: $\hat{\Gamma} = +0.00021 \pm 0.00015$, Holm $2.1{\times}10^{-4}$) and two negative-deficit cells with the predicted inversion. The single miss (12%: $\hat{\Gamma}$ marginally negative, arm table flat) is scored against the rule under its strong form and for it under its weak form; Section 5 carries the full accounting. The paying threshold *relocated* across eras (15% ⟶ 10%) as crisis density shifted—the fragility replicates in pattern, not in location, so any deployment must estimate the window's location for its own era.

The NBER cell: temporal custody. The one cell whose ex-ante status a third party can certify used a labeler with no free threshold anywhere: regime = NBER recession state × 50-day trend, the causal primary variant admitting a recession only from its *announcement* date. The registration was deposited with the Open Science Framework at 22:47 ET on 2026-07-15; NBER dates were first joined to the panel at 22:55 ET. Recession-cell dormancies are crisis-scale by construction (up to 2,557 trading days). The screen *fails* both variants ($z = 1.74$ primary)—the third dissociation of screen from value—while the deficit prices positive: $\hat{\Gamma} = +0.00081 \pm 0.00033$, strengthening to $+0.00105 \pm 0.00033$ net of 25 bps crisis-window costs, with the ordering exactly as the rule demands (Holm $1.6{\times}10^{-4}$ gross; the full retaining chain at Holm ≤ $3.1{\times}10^{-5}$ net). Disclosures at equal prominence: the lodged leave-one-out and era-blocked slices show the deficit survives every one of 20 single-event exclusions and all calendar exclusions (drop-2020: +0.00042 ± 0.00026; drop-2008/09: +0.00036 ± 0.00030), but its mass concentrates in the two deep-dormancy recession entries (January 2009, dormancy 1,378 days; June 2020, dormancy 2,445 days; dropping the former alone cuts $\hat{\Gamma}$ to 41% of headline), and the 2010s era block—which contains zero recession-cell reactivations—is negative-significant. The cell is event-robust in the strict sense, two-recession-concentrated, and not era-consistent; it is materially weaker than the drawdown cell's robustness profile, and we say so rather than average it away.

4 Costs and economics

A retention effect that dies under transaction costs is a curiosity; this one strengthens. The pre-registered cost battery charges crisis-window effective costs of 25/50/100 bps under a both-pay convention (every arm, including the protected twin, charged its own turnover; an arm-only variant was lodged as sensitivity and matches to the digit, because the twin's cost term cancels in the regret difference).

The deficit is cost-amplified. On the positive-deficit equity cell, the net-of-cost deficit rises monotonically with the cost tier: $\hat{\Gamma}_{\text{net}} = +0.00126 \pm 0.00020$ at 25 bps, $+0.00161 \pm 0.00021$ at 50, $+0.00230 \pm 0.00024$ at 100, with the four-arm ordering intact at every tier and the headline contrast strengthening from Holm $4.6{\times}10^{-19}$ to $1.3{\times}10^{-29}$. The mechanism is turnover, and it is the economically interpretable one: a reactivated always-retrain learner churns its top-k book hardest exactly when spreads are widest (measured turnover 1.03/day for the deployed baseline versus 0.62 for the calmest retaining arm). Paying to remember buys, among other things, *not trading like a fresh learner during a crisis onset.* The same signature reproduced ex ante on the NBER cell (+0.00081 → +0.00105 net of 25 bps; turnover 1.04 vs. 0.63)—a labeler the cost battery never saw.

The inversion was a gross-only phenomenon. Under the volatility-band labeler, where $\hat{\Gamma} < 0$ and the always-retrain baseline won gross, the inversion *reverses* net of costs at every tier (paired difference -0.00075 ± 0.00010 at 25 bps, $p = 1.2\times10^{-16}$): net of realistic crisis-window costs the retaining arm wins even where the parameter-level deficit is negative, because the turnover discipline alone is worth more than the deficit costs. We report this as economics, not vindication: the sign rule prices the *parameter* claim, and the turnover product is separately purchasable—a no-trade hysteresis band on the deployed baseline is the obvious desk countermeasure, and a banded-baseline control battery is lodged as a forward registration with its adverse branch stated (if banding alone closes the net gap where $\hat{\Gamma} \le 0$, the net-cost sentence above is relabeled as buyable discipline, at full prominence). One sensitivity is disclosed and not acted on: charging costs inside the structure screen would flip the drawdown labeler's pooled sub-criterion from −0.10% to +3.0%; the screen as registered is a gross-metric criterion and remains a fail.

What the cost model omits is market impact, borrow, and capacity; every substrate outside the two equity batteries remains gross; and none of these numbers is a P&L claim (Section 6.2). For any future alpha claim the lodged protocol of record is deflated Sharpe [3] plus an SPA/Reality-Check family test [13, 34], per the backtest-overfitting literature [14, 20].

5 The sign rule: a lodged, once-resolved hypothesis

The record above compresses to one deployment rule: *the sign of* $\hat{\Gamma}$ *prices retention.* Positive deficit ⇒ the retaining arms collect it; zero or negative ⇒ retention pays premium without claims. This section states exactly how much evidence that rule has, in the register the evidence deserves.

The confession first. When first formulated, the rule was postdictive—written after seeing both 1990–2025 outcomes, and sharing its A1 arm with the contrasts it prices (its non-circular content is that the protected twin sides with the retaining arms exactly when the deficit is positive). The weak/strong scoring taxonomy below was formalized only after the withheld-era runs, and the rule as lodged was scoped to walk-forward designs; we decline to hide behind that scoping and count the stitched designs against ourselves. One further scoring choice is confessed rather than buried: no registration fixed whether the gross or the net-of-cost register adjudicates the NBER cell's strong form—gross, its retaining chain is partial; net (the economically primary register by the cost battery's own argument, an argument that predates the cell), it is a full hit—and the amendment lodged for the constituent test fixes the net register in advance so the ambiguity cannot recur.

The inclusive scorecard. Scoring every cell the rule could touch—six withheld-era cells, the two schedule-resampled (stitched) 1990–2025 designs, and the four NBER cells—the record is **8/12 under the strong form** (sign predicts the arm-table direction: ordering when significantly positive, inversion when negative) and **12/12 under the weak form** (dissociation iff $\hat{\Gamma}$ significantly positive). The four strong-form misses share one shape: a marginally significant $\hat{\Gamma}$ with a flat arm table. Dependence caveats at full prominence: the withheld 15% full-era cell is the union of its two sub-era cells; the four threshold cells share one history's crises; the stitched designs resample the same panels. The scorecard's evidential force is closer to two-to-three effectively independent observations, in both directions, than to twelve trials—and the two 1990–2025 walk-forward tests that motivated the rule are not counted at all.

Table 2: Provenance grades for the program's registrations (compact; the full ledger, with commit hashes and verification instructions, is released). OSF project ID anonymized for review. [TODO: replace OSF ID with anonymized view-only link]

Registration	Custody grade
Gates + dissociation; drawdown labeler; threshold sweep (2026-07-14)	self-attested (spec and results entered the repository together)
Costs, replay, seed parity; mechanism probes; withheld era (2026-07-15)	git-separated (spec pushed before runs; server timestamps)
NBER announce-lagged forward test	*resolved under third-party custody* (OSF 22:47 ET; data joined 22:55 ET; hit)
CRSP constituents + T-split amendment; temporal-deployment test	forward, third-party custody (open)

Custody. Because a rule of this shape is exactly the kind of thing a file drawer manufactures, every registration in the program is graded by the strength of its timestamp custody, against ourselves (Table 2). The early batteries are self-attested; the later batteries have git-separated custody (specification pushed publicly before results existed); the NBER cell resolved under third-party (OSF) custody; and all future registrations are OSF-first, which is the only available answer to the unreportable-frozen-specs question.

The live test. The rule's standing changes register only with temporally split measurement, so the program's next cells are already committed. (1) *The deployment-form temporal test*: all diagnostics frozen on data through **2026-06-30** ($\hat{\Gamma} = +0.00081 \pm 0.00033$ on the NBER cell; $+0.00091 \pm 0.00020$ on the drawdown cell—both positive, so the lodged prediction is that the retaining arms outperform in the outcome window), scored against realized reactivations in 2026-07-01 through 2027-03-31 on a pre-named date, **2027-04-07**, with a void declared (not a hit) if no qualifying reactivation occurs. One clean refuting cell falsifies the deployment form of the rule. (2) *Constituent-level replication*: the CRSP S&P 500 panel, registration lodged before any data access, with a T-split amendment fixing the net-cost register and the window estimator in advance—the test that decides whether the deficit survives disaggregation, the canonical worry for industry-level effects [10, 25]. Both are reported at headline prominence regardless of outcome.

6 Field kit and limitations

6.1 Running the protocol on your own book

(1) **Freeze the design.** Fix the labeler family, thresholds, probe length, dormancy cutoff, and test family in writing; lodge the document externally (OSF or equivalent) before any run. The record above is the argument that this step is load-bearing.
(2) **Label causally.** Regime labels must be computable from information through $t-1$—announcement dates, not revision dates, for any official indicator.
(3) **Run the screen.** One model per regime versus the same machinery on ≥ 50 block-shuffled label draws, on *your decision metric*, walk-forward, with a split-half check. A fail predicts your regime-aware evaluation should go flat; run it anyway and distrust everything if it does not.
(4) **Measure $\hat{\Gamma}$.** Two arms: your deployed allocation policy, and its twin with per-regime parameters pinned through dormancy. $\hat{\Gamma}$ = probe-window regret difference. Charge your own crisis-window cost schedule (both-pay); keep $\hat{\Gamma}_{\text{net}}$.
(5) **Price it.** Apply the boxed rule of Section 2.4 with your sleeve size, measured dormancy scale, and honest carrying cost.
(6) **Decide, and re-estimate.** $\hat{\Gamma}_{\text{net}}$ significantly positive and the rule satisfied: retention pays, and any isolation-preserving implementation collects it. Otherwise: do not buy retention machinery this era—and re-run the diagnostic when crisis density shifts, because the paying window's location moved across eras in our record.

6.2 Limitations, register, and what we do not claim

Single histories. Every real-data cell is one realized market history; seed intervals quantify implementation noise conditional on that history. The record spans 1926–2025 and scores in both directions, but it is three walk-forward windows and one country.

Aggregation. The equity substrate is industry portfolios; aggregation itself can manufacture slow dynamics [19], crisis-conditional behavior is a documented feature of this territory [4, 5, 15, 32], and industry-level effects need not survive disaggregation [10]. The lodged CRSP constituent test is the decider, and until it resolves, the deficit is a fact about the aggregate cross-section.

One labeler family, one era-dependent window. The positive deficits live under drawdown- and recession-type labelers with slow crisis dynamics; the paying threshold relocated across eras. The protocol ships with the instruction to estimate the window locally, not with a universal constant.

Mechanism. On real data the measured deficit is a persistent allocation-level service gap, not demonstrated forgetting; the controlled dissociation of mechanisms lives in the companion [2], and buying our diagnostic does not require buying any mechanism story.

No alpha. Every number here is decision regret at portfolio level, gross except where a modeled crisis-window cost schedule is stated; nothing is a claim of live trading profitability [23]. The protocol's product is a decision—pay to remember, or don't—and its warranty is the record of Table 1: three flats, two inversions, two orderings, all published under the same rule.

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